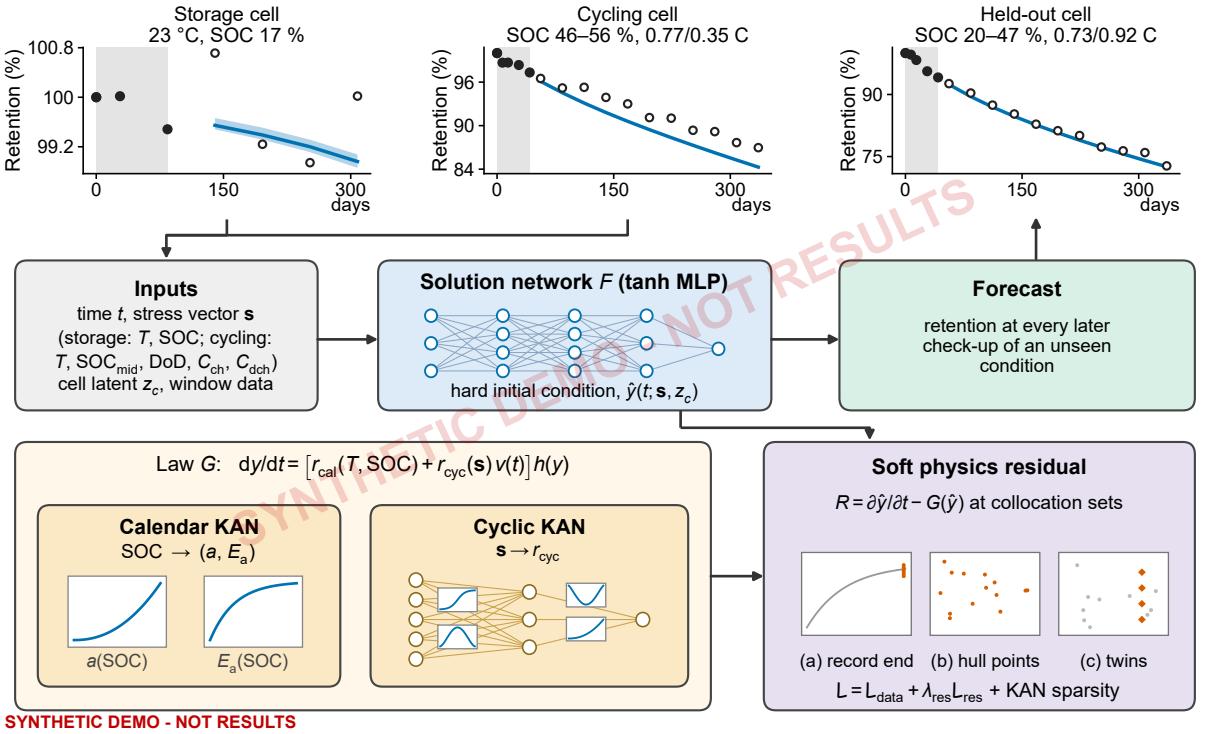

Storage cell
23 °C, SOC 17 %
Retention (%)
100.8
100
99.2
0
150
300
days
Cycling cell
SOC 46–56 %, 0.77/0.35 C
Retention (%)
96
90
84
0
150
300
days
Held-out cell
SOC 20–47 %, 0.73/0.92 C
Retention (%)
90
75
0
150
300
days
Inputs
time t, stress vector s
(storage: T, SOC; cycling:
T, SOC_mid, DoD, C_ch, C_dch)
cell latent z_c, window data
Solution network F (tanh MLP)
hard initial condition, ŷ(t; s, z_c)
Forecast
retention at every later
check-up of an unseen
condition
Law G: dy/dt = [r_cal(T, SOC) + r_cyc(s) v(t)] h(y)
Calendar KAN
SOC → (a, E_a)
a(SOC)
E_a(SOC)
Cyclic KAN
s → r_cyc
Soft physics residual
R = ∂ŷ/∂t − G(ŷ) at collocation sets
(a) record end
(b) hull points
(c) twins
L = L_data + λ_res L_res + KAN sparsity
SYNTHETIC DEMO - NOT RESULTS
SYNTHETIC DEMO - NOT RESULTS